

FPSO PROJECT PROJECT No. 20171

Approved Suppliers List

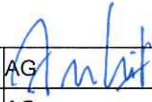
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1. INTRODUCTION

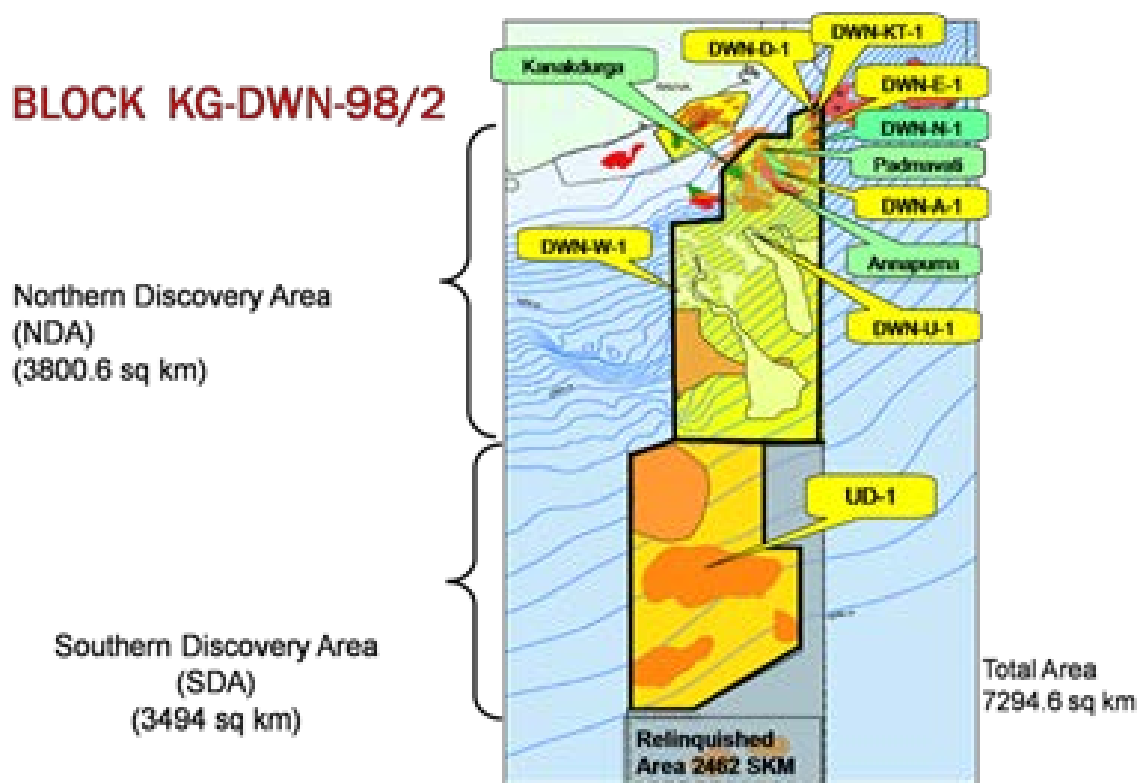
1.1 PROJECT BACKGROUND

ONGC intends to develop the New Exploration Licensing Policy (NELP) Block KG-DWN-98/2, in deep waters off the east coast of India. The discovery areas in the block have been categorized as Northern Discovery Area (NDA) and Southern Discovery Area (SDA). The scope of the present development is limited to Cluster 2 of NDA only.

A location map identifying the KG-DWN 98/2 development area is presented as Figure 1.1.

The KG-DWN 98/2 block is located within 25-40 km of the nearest land-fall and is adjacent to RIL's KG-D6 field and GSPC's Deendayal field. The water depth in this block varies between 300 metres and 3,200 metres.

Figure 1.1: Block KG-DWN-98/2 Development Area



The fields discovered in the area comprise of a number of oil and gas discoveries and are spread over a large area. Each field has technical challenges such as high wax, hydrates and sand production.

The Pre-FEED assumes NDA covers the following production area:

- Cluster 2
 - Mix of oil and gas, located in the south of NDA, includes the following fields:
 - Oil fields (Cluster 2A) – DWN-A-2; DWN-P-1; DWN-M-3, DWN-M-1; DWN-M-4 and DWN- G-2-2.

- Gas fields (Cluster 2B) – DWN-R-1; DWN-U-3; DWN-U-1 and DWN-A.

ONGC is targeting First Gas from the KG-DWN-98/2 U-field by June 2019 via utilization of the Vashishta infrastructure.

The Vashishta and S-1 fields are located adjacent to the KG-DWN-98/2 field in a water depth range of 500 to 700 meters.

In 2017, construction on Vashishta and S-1 field development is ongoing. The Vashishta and S-1 field development is comprised of three wells. Two wells, VA-DA and VA-DB are located in the Vashishta field and one well, S1-AC at the S-1 field will be tied into the Vashishta flowline system. The S1-AC well is located in a water depth of approximately 412 meters. Dual subsea flowlines will be installed to the Vashishta onshore terminal at Odalarevu, adjacent to the existing G1 and GS15 onshore terminal.

A phased development is considered as recommended in the FDP:

- Phase 1: Cluster 2B (gas); and
- Phase 2: Cluster 2A (oil and water injection)

Processing of raw well stream oil from the Cluster 2A wells will be undertaken on an FPSO as follows:

- Oil will be stabilized and stored in hull tanks prior to transfer at sea to an offloading tanker via a tandem offloading hose.
- Associated gas will be processed and exported to the CPP platform via a subsea pipeline, subsequently transferring to the Onshore Gas Terminal. Both hydrocarbon and water dew pointing operations will be undertaken to avoid liquid dropout in the subsea pipeline.
- Produced water will be treated and disposed of overboard.

1.2 DEFINITIONS

COMPANY	Oil and Natural Gas Corporation Limited (ONGC)
CONTRACTOR	Shapoorji Pallonji Bumi Armada Godavari Private Limited (SPBAG)
PROJECT	Charter Hiring of FPSO for NELP Block KG – DWN 98/2 Development Project Cluster - II
Services	Any service or work performed by Supplier that must be performed to comply with the requisition requirements, or the contract, to procure, design, manufacture, and deliver the work.
Standards	Industry Codes, Standards, Guides, and Recommended Practices referenced herein. Meaning the latest issue or edition in force at the end of the Supplier bid validity date or the contract date.
Supplier	<u>At quote stage</u>: any entity invited to provide a quotation for the equipment and/or any SubCONTRACTORS thereto. <u>At Purchase stage</u>: any entity contracted for the supply of the equipment and/or any SubCONTRACTORS thereto. In all cases, the Supplier / CONTRACTOR is responsible for performance of all Work and will be the single point of contact for all Work related issues. The CONTRACTOR will not receive information from nor respond directly to Sub-suppliers / SubCONTRACTORS

1.3 SCOPE OF DOCUMENT

This document states the list of major equipment suppliers, any of which shall be used by the CONTRACTOR for the afore mentioned project.

1.4 Major Equipment Supplier

Sr. No	Vendor Name	Country	Remarks
1.	Power Generation System		
1. a	Siemens	Norway	
1. b	General Electric (GE)	USA	
2.	E-house & Integrated Control & Safety System		
2. a	Siemens Pte Ltd	Singapore	
2. b	ABB Pte Ltd	Singapore	
2. c	General Electric (GE)	Singapore	
2. d	Yokogawa (ICSS)	India / Singapore	
2. e	Emerson (ICSS)	India / Singapore	
3.	Booster Export Compressor and Gas Lift Compressors		
3. a	Siemens (Dresser Rand)	France	
3. b	General Electric (GE)	Italy	
3. c	Man Turbo	Germany	
4.	Flash Gas Compressor		
4. a	Kobelco	India / Japan	
4. b	Man Turbo	Germany	
5.	Flare Gas Compressor		
5. a	Gardner Denver Nash	Singapore	
5. b	Flowserve	Germany	
5. c	Man Turbo	Germany	
5. d	Howden	UK	
5. e	Kobelco	Japan/USA	
5. f	Mayekawa	USA	
5. g	Dresser Rand	USA	
5. h	Neuman Esser	Germany	
5. i	Ro Flo Compressors LLC	USA	
5. j	GEA	Germany	
6.	Sea Water Lift Pump		
6. a	Flowserve (Austria) GmbH	Austria	
6. b	Framo AS	Norway	
6. c	KSB Aktiengesellschaft	Germany	

6. d	Sulzer Pump	UK / Brazil / India	
6. e	Clyde Union	Canada	
6. f	Peerless	Australia	
6. g	Hamworthy (Wartsila)	Singapore	
7.	Separation system		
7. a	Frames B.V.	Netherlands	
7. b	VME Process Ltd.	Indonesia / Malaysia/Singapore	
7. c	National Oilwell Varco	Norway / Malaysia	
7. d	Cameron	USA/ Malaysia	
7. e	CDS Separation System	UK	
7. f	Zeta - PDM	UK / Malaysia	
8.	Sea Water Injection Pumps		
8. a	Sulzer Pump	UK	
8. b	SPX Clyde Union	UK	
8. c	Flowserve Corporation	USA	
8. d	Framo AS	Norway	
8. e	Nuovo Pignone S.p.A (GE Group)	Italy	
8. f	Ebara Pumps	Japan	
8. g	DMW Corporation	Japan	
8. h	KSB	Germany	
8. i	Weir	Italy	
9.	Sea Water Treatment Package + Sulphate Reduction Package		
9.a	VWS Westgarth Ltd (Veolia)	UK	
9.b	National Oilwell Varco / Fjords	Norway / Malaysia	
9.c	VME Process Ltd.	Indonesia / Malaysia/Singapore	
9.d	Jord Oil & Gas Systems B.V.	Netherlands	
9.e	Cameron Process	Singapore	
10.	TEG Regeneration Package + Hydro Carbon Dew Point Package		
10. a	VME Process Ltd.	Indonesia / Malaysia/Singapore	
10. b	Suez	Australia / Singapore	
10. c	National Oilwell Varco / Fjords	Norway / Malaysia	
10. d	NATCO Japan Co. Ltd.	Japan	
10. e	Global Process Systems Sdn Bhd	Malaysia	
10. f	Process Group	Australia	
10. g	Frames Process Systems, BV	Netherlands	
10.h	Jord Oil & Gas Systems B.V.	Netherlands	

11.	Crude Transfer Pump (Centrifugal Pump)		
11.a	Sulzer Pump	UK / Brazil	
11.b	SPX Clyde Union	UK	
11.c	Flowserve Corporation	USA	
11.d	DMW Corporation	Japan	
11.e	Framo AS	Norway	
11.f	Nuovo Pignone S.p.A (GE Group co)	Italy	
11.g	KSB	Germany	
12.	Crude and Gas Metering System		
12.a	Krohne Oil and Gas	Netherlands / Malaysia	
12.b	Alderly Systems Ltd	UAE	
12.c	Oval Asia	Singapore	
12.d	Emerson	Singapore	
13.	Pedestal Cranes		
13.a	Favelle Favco Cranes	Malaysia	
13.b	Seatrax	USA	
13.c	Palfinger Dreggen AS	Norway	
13.d	National Oil Well Varco	USA	
13.e	Italgru Srl	Italy	
13.f	Liebherr-MCCtec Rostock GmbH	Germany	
14.	Firewater Pump		
14. a	Wartsila Pumps Pte Ltd	Norway / Singapore	
14. b	Sulzer Pump	UK / Brazil	
14. c	Framo AS	Norway	
14. d	Nuovo Pignone S.p.A (GE Group)	Italy	
14. e	KSB	Germany	
14. f	Pon Power	Netherland	
14. g	Flowserve Corporation	USA	
14. h	SPP Pumps Ltd	UK	
14. i	Ellehammer A/S	Norway	
14. j	Fluid Power Equipment	USA	
14. k	Iron Pump A/S	Denmark	
14. l	Peerless Australia Pty Ltd	Australia	
15.	Turret Mooring System (including Swivels)		
15. a	National Oilwell Varco (APL)	Norway	
15. b	Bumi Armada Berhad	Malaysia	

15. c	OneSubsea (Framo AS)	Norway	
16.	Mooring Chains		
16. a	Franklin Offshore (Jiangsu Asian Star)	Singapore	
16. b	Ramnas Bruk	Sweden	
16. c	Sanmar Chain	Singapore	
16. d	Vicinay Cadenas SA	Spain	
16. e	Zheng Mao Group Co,Ltd	China	
16. f	Jiangsu Asian Star	China	
16. g	Hamanaka Chain	Japan	
16. h	Mooreast Asia	Singapore	
17.	Mooring Wires		
17.a	Franklin Offshore	Singapore	
17.b	Oceanic Offshore Engineering	Singapore	
17.c	Rulong Group	China	
17.d	Usha Martin	India	
17.e	Bridon International	United Kingdom	
17.f	Redaeilli	Italy	
17.g	Trefileurope	France	
17.h	Scanrope	Norway	
17.i	Jiangsu Asian Star	China	
17.j	KTL Offshore	Singapore	
17.k	Mooreast Asia	Singapore	
18.	FPSO Conversion and Refurbishment-Life Extension Services		
18.a	Jurong Shipyard Ltd.	Singapore	
18.b	Keppel Shipyard Ltd.	Singapore	
18.c	Drydocks World Dubai LLC	Dubai	
19.	Topsides Module Fabricators		
19.a	Jurong Shipyard Ltd. (Sembcorp)	Singapore	
19.b	Keppel Shipyard Ltd.	Singapore	
19.c	National Oilwell Varco (PT Profab)	Indonesia	
19.d	Dyna-Mac Engineering Services Pte Ltd	Singapore	
19.e	Siemens Pte Ltd.	Indonesia / Singapore	
19.f	PetroVietnam Technical Services	Vietnam	
19.g	Aibel Offshore	Thailand	
19.h	VWS Westgarth Ltd (Veolia)	UK	

19.i	Jord Oil & Gas Systems B.V.	Netherlands	
19. j	VME Process Ltd.	Indonesia / Malaysia/Singapore	
19. k	Suez Oil & Gas Systems	Singapore	
19.l	MecPec Marine & Offshore Engineering Pte Ltd	Singapore	
20.	Transportation & Installation Services Providers for FPSO Installation		
20.a	Boskalis Marine Offshore Contracting BV	Netherlands / Singapore	
20.b	InterMoor Asia	Singapore	
20.c	Sapura Offshore Sdn Bhd	Malaysia	
20.d	Subsea7 Australia Contracting Pty Ltd	Australia	
20.e	Franklin Offshore	Singapore	
20.f	Saipem Malaysia	Malaysia	
20.g	McDermott Asia	USA / Malaysia	
21.	Electrostatic Coalescer		
21.a	Fjords Processing		
21.b	VME Process Ltd.	Indonesia / Malaysia/Singapore	
21.c	Frames Separation	Netherlands / Malaysia	
21.d	Bea Technologies Spa	Italy	
21.e	Sulzer Chemtech AG	Switzerland	
21.f	Cameron Process Systems	Singapore	
21.g	Suez Oil & Gas Systems	Singapore	
21.h	Kanfa Group	Norway	
21. i	MacKenzie Hydro Carbons	Australia	
21. j	Prosernet	France / Malaysia	
22.	Boilers		
22.a	Greens Power	UK	
22.b	Harris Pye	UAE	
22.c	Mitsubishi Heavy Industries	Japan	
22.d	Alfa Laval	Singapore	
22.e	Kawasaki Heavy Industries	Japan	
22.f	Saacke Marine Systems	Norway	
22.g	Hamworthy Combustion	UK	
23.	Offloading Hose, Reel and Offloading System		
23.a	SCANA Offshore AS	Norway	
23.b	Trellerborg Industries	Australia	
23.c	Techflow Marine	UK/Singapore	
23.d	IHC Asia Pacific Pvt Ltd	Singapore	

23.e	Timberland Equipment Ltd	Canada	
23.f	MacGregor Norway AS	Norway	
23.g	NOV Completions & Production Solutions	Singapore	
24.	Produced Water Treatment		
24.a	JORD Oil and Gas Systems B.V	Netherlands / Singapore	
24.b	NOV Process & Flow Technologies	Singapore / Batam	
24.c	VME Process	Malaysia/ Batam	
24.d	Process Group	Australia	
24.e	Frames Group	Malaysia	
24.f	Prosep	Malaysia	
25.	Process Pumps (Centrifugal, Diaphragm, Piston, Axial, High Pressure)		
25.a	Sulzer		
25.b	Weir	Germany / Malaysia	
25.c	Flowserv	Switzerland / Malaysia	
25.d	Ebara Corporation	Japan	
25.e	Arai Pump	Japan	
25.f	SPX Flow	UK / Singapore	
25.g	Iron Pump	Denmark	
25.h	Framo AS	Norway	
25.i	KSB	Malaysia	
25.j	Wartsila Pump	Singapore	
25.k	SPP Pumps	UK	
25.l	Fluid Power Equipment	USA	
25.m	Peroni Pompe	Italy	
25.n	Viking Pump	USA	
25.o	DMW Corporation	Japan	
25.p	RAM Pumps	UK	
25.q	CP Pumps & Systems	UAE	
25.r	Goulds Pump	USA	
25.s	Peerless	Australia	
26.	Diesel Generators		
26.a	Cummins	Singapore	
26.b	Pon Power	Singapore	
26.c	Tractors (Caterpillar)	Singapore	
26.d	Wartsila	Finland	
27.	Topside & Marine Valves (Ball, Globe, Gate, Check, Butterfly)		
27.a	WEIR Valves & Controls Ltd.	UK	

27.b	Jordan Valve	USA	
27.c	Valvitalia	Italy	
27.d	Virgo Valves	India	
27.e	Severn Glocon India Private Ltd	India	
27.f	Flowserve Corporation, Flow Control Division	Germany	
27.g	Emerson Process Management	USA	
27.h	Coyard - Saccap	France	
27.i	Petrol valves S.r.l	Italy	
27.j	L&T Valves Ltd	India	
27.k	Orseal Ltd, Bolton	UK	
27.l	Hawa Valves	India	
27.m	Bonney Forge	Italy	
27.n	Ace Valves	South Korea	
27.o	Mir Valve	Malaysia	
27.p	KVC Valves	Japan / South Korea / UK	
27.q	Neway	China	
27.r	Kitz	Japan	
27.s	George Fischer	Switzerland	
27.t	IMI CCI	Korea	
27.u	Galperti	Italy	
27.v	SS Valve	Korea	
28.	Instrument Valves (Double Block Bleed, Needle, Ball, Check)		
28.a	Swagelok	USA	
28.b	Niton Valve Industries	India	
28.c	Friulco S.p.A	Italy	
28.d	Galperti	Italy	
28.e	Oliver Valves	UK	
28.f	Douglas Chero S.p.a	Italy	
28.g	Firsa S.p.a	Italy	
28.h	Fisher Controls	USA	
28.i	KOSO	India	
28.j	Dresser Valve	India	
28.k	Parker	USA	
28.l	Kitz	Japan	
28.m	Flowserve Corporation, Flow Control Division	Germany	
28.n	Emerson Process Management	USA	
28.o	Coyard -Saccap	France	
29.	Actuators		
29.a	Bettis	USA	
29.b	Rotork	UK	

29.c	Biffi Italia	Italy	
29.d	Emerson Process	USA	
29.e	Flowserve Corporation, Flow Control Division	Germany	
29.f	Flo - Tork	USA	
29.g	AUMA RIESTER	Germany	
29.h	Cameron	USA / Italy	
29.i	ATC	Netherlands	
29.j	Imtec Controls	UK	
30.	GRE Fittings		
30.a	Future Pipe Industries	Netherlands	
30.b	NOV FGS	Singapore	
30.c	Fiberbond	USA	
30.d	Smith Fiberglass	USA	
30.e	Conley Corp	USA	
30.f	Bond Strand	India	
31.	Heat Exchangers – Shell & Tube		
31.a	Alfa Laval Olmi S.p.A	Italy	
31.b	Heat Exchangers Specialists	Singapore	
31.c	Sterling Thermal Technology Ltd.	UK	
31.d	FBM Hudson Italiana SpA	Italy	
32.	Heat Exchangers – Plate & Frame		
32.a	Tranter	USA	
32.b	HISAKA	Japan	
32.c	Alfa Laval Olmi S.p.A	Italy/Malaysia/Singapore	
32.d	Sondex	USA	
32.e	APV	USA	
32.f	GEA Ecoflex	Malaysia	
33.	Heat Exchangers – Printed Circuit		
33.a	Heatric	UK	
33.b	Heat Exchangers Specialists	Singapore	
33.c	Alfa Laval Olmi S.p.A	Italy	
33.d	Tanktech	Korea	
34.	Chemical Injection System – Packaged Item		
34.a	VME Process	USA/Singapore/Malaysia	
34.b	Geveke Pompen	Italy	
34.c	GlobalTech Systems Engineering Pte Ltd	Singapore	
34.d	Petronash FZE	UAE	

34.a	LEWA (Nikkiso)	Japan/Singapore	
35.	Pressure Vessels		
35.a	Petra Fabricators Sdn Bhd	Malaysia	
35.b	Amalgamated Metal Corporation	Malaysia	
35.c	Jutasama Sdn. Bhd.	Malaysia	
35.d	Vesseltech Engineering Sdn. Bhd.	Malaysia	
35.e	Tech Offshore Marine	Singapore	
35.f	Frames Process Systems	Nederlands/Malaysia	
35.g	MSET Engineering Corporation	Malaysia	